

Introduction

The diffraction pattern of a Helix is an interesting pattern that is formed as its structure resembles two thin wires kept at an angle to each other and what is observed is similar to the combined diffraction pattern for individual pair of wires.

We observe two different lines forming on the screen at an angle, and within each of them we see a finer and a coarse diffraction pattern. The finer pattern is formed by diffraction around two parallel parts of the wire, while the coarse pattern is formed by diffraction by a single wire.

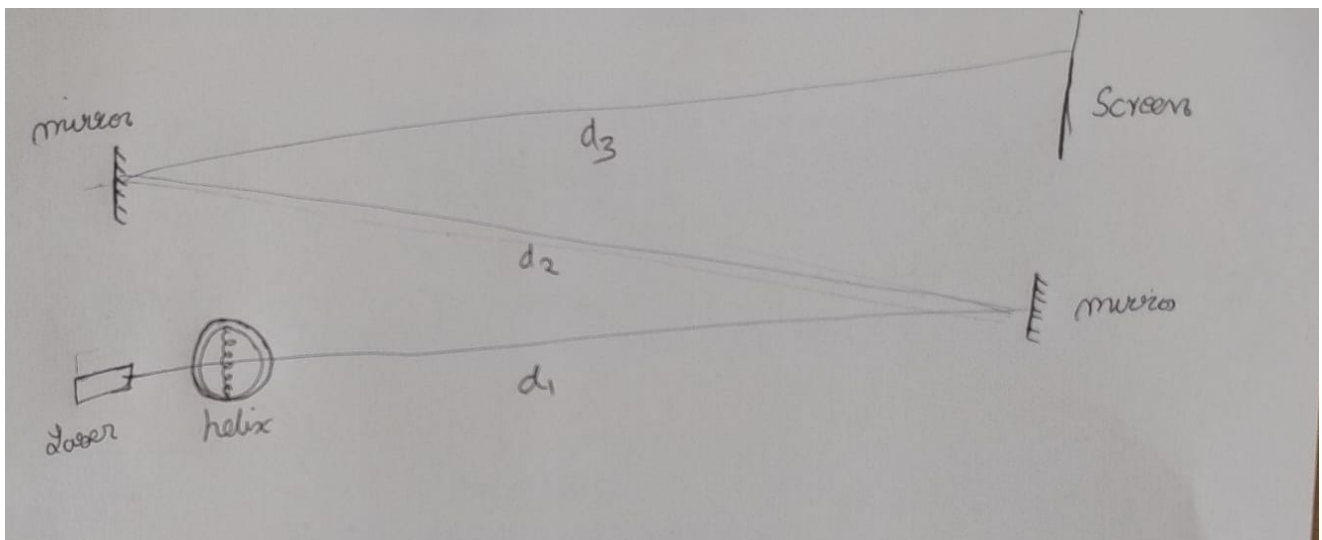
Thus by studying the diffraction pattern for the helix, we can determine different parameters of the helix like the thickness of the wire forming the helix, pitch of the helix and its diameter. The purpose of the experiment is to determine these parameters.

Experiment Setup and Procedure

Apparatus

A laser of known wavelength, helix, two mirrors, screen. stands for laser and the helix.

Experiment Setup



The purpose of mirrors is to extend the distance between the screen and the source. Given the dimensions of the spring, the diffraction angles for the dark and bright spots are very small. So a large distance is required to reliably measure those angles.

Procedure

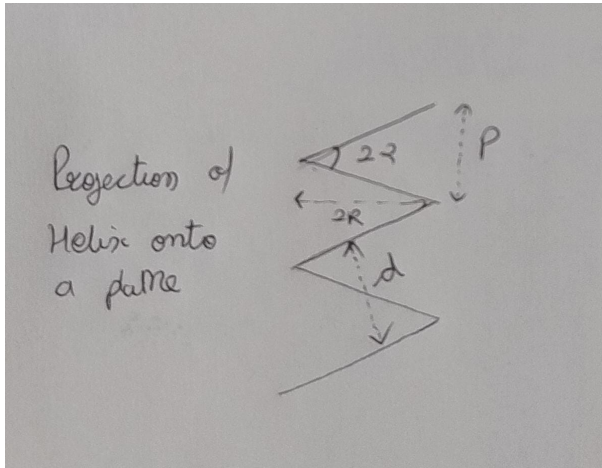
After the setting up the experiment and adjusting the mirrors so that the diffraction pattern forms on the screen, using a paper attached to the screen, dark spots formed by the fine

and coarse diffraction are marked separately and the distance between them is measured. The angle between the two lines is measured.

Using a measuring tape, the distances between source, mirrors and screen are measured. Adding them together, we get the total distance travelled by the light between the source and the screen.

Using the wavelength of the laser source and this data, the parameters of the spring are calculated.

Formulae and Relations



D = distance between source and the screen. = $d_1 + d_2 + d_3$

$\lambda = 650 \text{ nm}$

Angle between the lines on screen = 2α

t = thickness of the wire = $\frac{\lambda D}{\Delta x_c}$ where

Δx_c = spacing of the coarse diffraction pattern.

$d = \frac{\lambda D}{\Delta x_f}$ where

Δx_f = spacing of the fine diffraction pattern.

$P = \text{Pitch} = \frac{d}{\cos \alpha}$

$R = \text{Radius} = \frac{P}{2\pi \tan \alpha}$

Data and Analysis

The following measurements were taken.

$D = 273 \text{ cm}$

$2\alpha = 22^\circ$

$\therefore \alpha = 11^\circ$

$\Delta x_c = 1.1 \text{ cm}$

$\Delta x_f = 2.2 \text{ mm}$

So, we get

$$d = \frac{\lambda D}{\Delta x_f} = 0.81 \text{ mm}$$

$$P = \frac{d}{\cos \alpha} = 0.825 \text{ mm}$$

$$R = \frac{P}{2\pi \tan \alpha} = 0.675 \text{ mm}$$

$$t = \frac{\lambda D}{\Delta x_c} = 0.16 \text{ mm}$$

Thus all the required parameters have been determined.